Syllabus for ITR

*Artificial Intelligence & Machine Learning *

Topics Covered

- Introduction to AI & Python for Data Science.
- Data Preprocessing & Feature Engineering.
- Supervised Learning – Regression.
- Supervised Learning – Classification.
- Unsupervised Learning.
- Neural Networks & Deep Learning Introduction.
- Deep Learning – Convolutional Neural Networks (CNN).
- Natural Language Processing (NLP).
- Reinforcement Learning & AI Applications.
- AI Model Deployment & Optimization.
- AI Ethics & MLOps.
- Capstone Project & Industry Use Cases.

1) Introduction to AI & Python for Data Science

- What is AI? History & Applications
- Types of AI (Narrow, General, Super AI)
- Machine Learning vs. Deep Learning vs. Data Science
- Overview of Supervised, Unsupervised, and Reinforcement Learning
- Setting up the AI/ML environment (Anaconda, Jupyter, Google Colab)

2) Python for AI & ML

- Python Basics (Data Types, Loops, Functions)
- Numpy & Pandas for Data Handling

- Matplotlib & Seaborn for Data Visualization
- Introduction to Scikit-Learn
- Exception Handling

3) Data Preprocessing & Feature Engineering

- Handling Missing Data & Data Cleaning
- Feature Scaling (Normalization & Standardization)
- Encoding Categorical Variables (One-Hot Encoding, Label Encoding)
- Feature Selection Techniques

4) Supervised Learning – Regression Models

- ILinear Regression & Polynomial Regression
- Multiple Linear Regression
- Decision Tree & Random Forest Regression
- Evaluating Regression Models (RMSE, R2 Score)

5) Supervised Learning – Classification Models

- Logistic Regression
- K-Nearest Neighbors (KNN)
- Decision Trees & Random Forest Classifier
- Support Vector Machines (SVM)

6) Unsupervised Learning & Clustering

- K-Means Clustering
- Hierarchical Clustering

- Principal Component Analysis (PCA) for Dimensionality Reduction

7) Neural Networks & Deep Learning Introduction

- Basics of Artificial Neural Networks (ANN)
- Activation Functions (ReLU, Sigmoid, Softmax)
- Forward & Backpropagation
- Introduction to TensorFlow & Keras

8) Deep Learning – Convolutional Neural Networks (CNN)

- Understanding CNN Architecture
- Building CNN models using TensorFlow/Keras
- Image Classification with CNN

9) Natural Language Processing (NLP)

- Tokenization, Stemming, Lemmatization
- Bag of Words & TF-IDF
- Sentiment Analysis with NLP
- Using Pre-trained Models like BERT

10) Reinforcement Learning & AI Applications

- Markov Decision Process (MDP)
- Q-Learning & Deep Q Networks (DQN)
- OpenAI Gym Environments
- AI in Gaming (Building a simple RL-based bot)

11) AI Model Deployment & Optimization

- Model Evaluation & Hyperparameter Tuning

- Deploying AI/ML Models using Flask & FastAPI
- Introduction to MLOps & Model Monitoring

11) Capstone Project & Industry Use Cases

- Project Options:
  1. Real-time Object Detection (YOLO/OpenCV)
  2. Chatbot (NLP + Transformers)
  3. Stock Price Prediction (LSTMs)
  4. Recommendation System (Collaborative Filtering)
- End-to-End ML Project (Choose between Image Classification, NLP, or Time-Series Forecasting)
- Resume & Interview Preparation for AI/ML Roles
- Understanding AI Ethics & Future Trends